Evaluating Unsupervised Discovery of Persona-Relevant Directions with Matryoshka Sparse Autoencoders

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Working paper summary · `autolabs-ebon.vercel.app`

Research question. Can an SAE identify coherent persona-relevant directions without a prespecified target trait? I investigate the discovery question raised in the limitations section of Chen et al. (2025). Prompt reachability was omitted from the final run; ease of prompt elicitation is reserved for future work.

Method. I train a Matryoshka BatchTopK SAE on 100M assistant-position tokens from `Qwen2.5-7B-Instruct` at decoder layer 19 (zero-based). It has 32,768 features, approximately 235M parameters, held-out FVE 0.717, and mean $L_0 = 40$. Candidate selection uses assistant-specificity, density-quantile, and prompt-informed arms. Both signs are steered across 24 scenarios, alongside 20 random directions and three supervised persona-vector controls. A blinded language-model judge describes selected steered-baseline pairs.

Recorded outcomes. The diagnostic criterion G_0 combines residual separability and cross-scenario consistency. It is satisfied by 60/128 features and two of three controls. Judge evaluation covers 40 selected signed feature directions and three random directions. Ten signed candidates receive consistent speaker-related descriptions above the naming-null threshold; all ten also pass G_0. They represent **nine distinct features**, because both signs of feature 2036 are named.

Selection arm	Features screened	G_0 features	Named signs
Assistant-specificity	48	25	2
Density-quantile	32	10	0
Prompt-shift	48	25	8

Interpretation. The two candidates selected without persona prompts are confounded by degraded output. Feature 148+ produces only punctuation in all 24 scenarios. Its "engaged and supportive" label describes the baseline response in all 48 judge comparisons, not the steered output. The coherence proxy nevertheless accepts 23/24 responses, exposing a validity problem. Feature 717+, labelled "composed and confident," passes that proxy on only 8/24 scenarios and also shows substantial degradation. The eight other named directions come from prompt-informed selection. The current evidence therefore does not establish coherent unlabelled persona discovery.

Limitations and next research steps. The study uses one model and layer, selective judge coverage, unmatched random perturbation norms, and three naming nulls selected for low separability. Reconstruction replacement yields 78% argmax agreement, below the configured 90% threshold. Future evaluation should require substantive coherent output, retain judgement polarity, use comparable nulls, and include independent human validation. Prompt-elicitation difficulty should be measured after coherent candidate effects have been established.

Reference: R. Chen, A. Arditi, H. Sleight, O. Evans, and J. Lindsey. *Persona Vectors: Monitoring and Controlling Character Traits in Language Models.* arXiv:2507.21509 (2025).

Artifacts: Code and full paper; SAE checkpoint. Final screen: `persona-3c-ded5477e`.